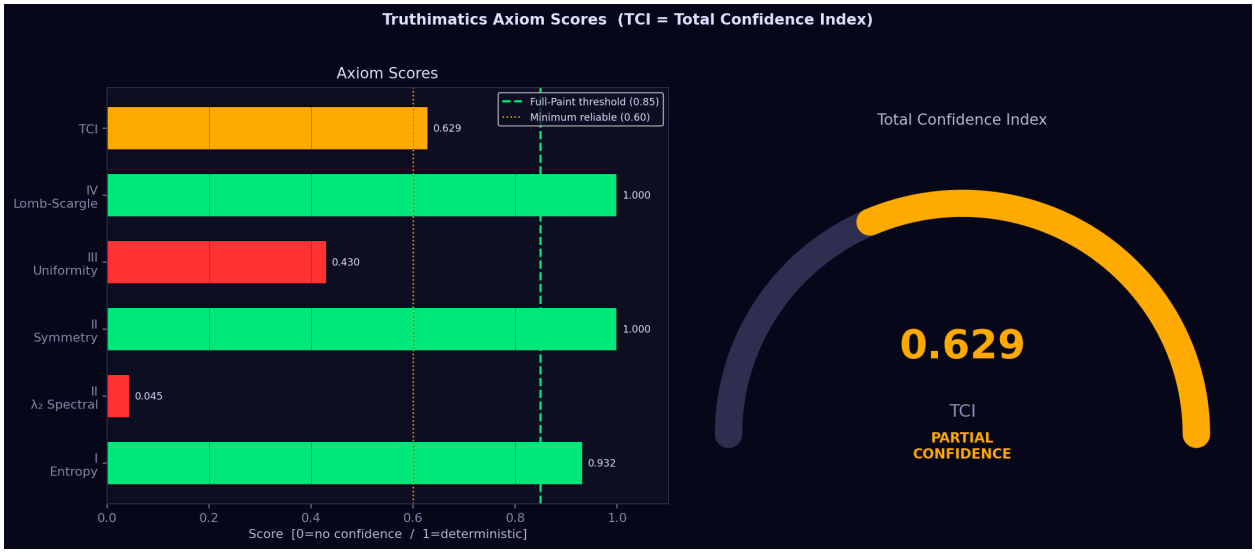
AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_32C8C68F37
Taxa	Tyrannosaurus
Clade	Theropod
PBDB Records	86
Mass (regression)	5728.6 kg (Campione & Evans 2012)
Mass (volumetric)	4393.8 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	5.169 m³
Validity Score	1.0000 / 1.0000
TCI	0.6292 (partial)
Axiom I Entropy	0.9321
Axiom II λ■	0.1802 (symmetry 100%)
Axiom III Uniformity	0.4305
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	60.0% Measured / 40.0% Inferred
Elements	25
CoG (m)	[0.388, 1.283, 0.0]
Generated	2026-04-30 23:51 UTC

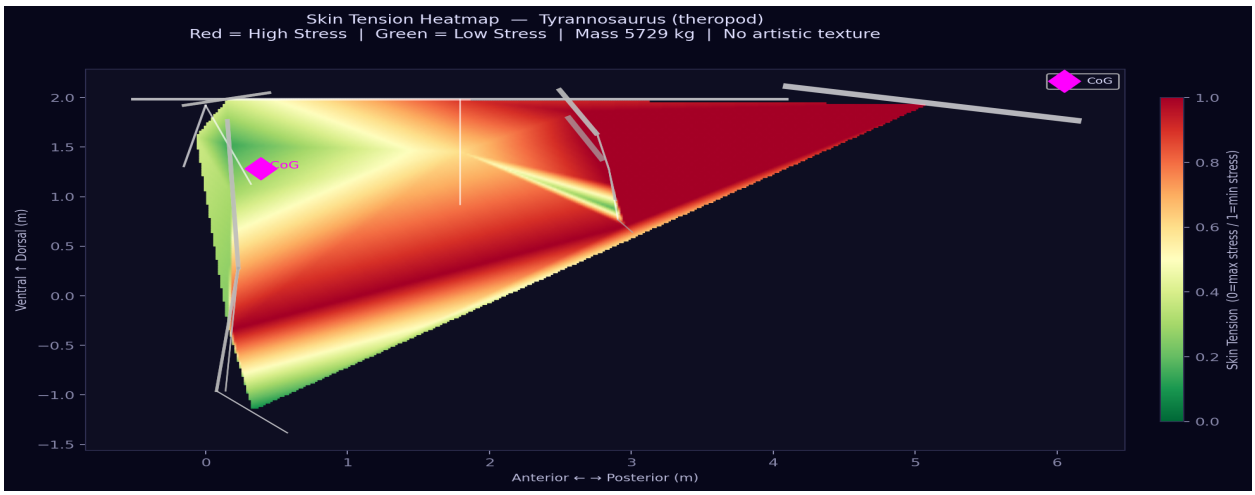
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



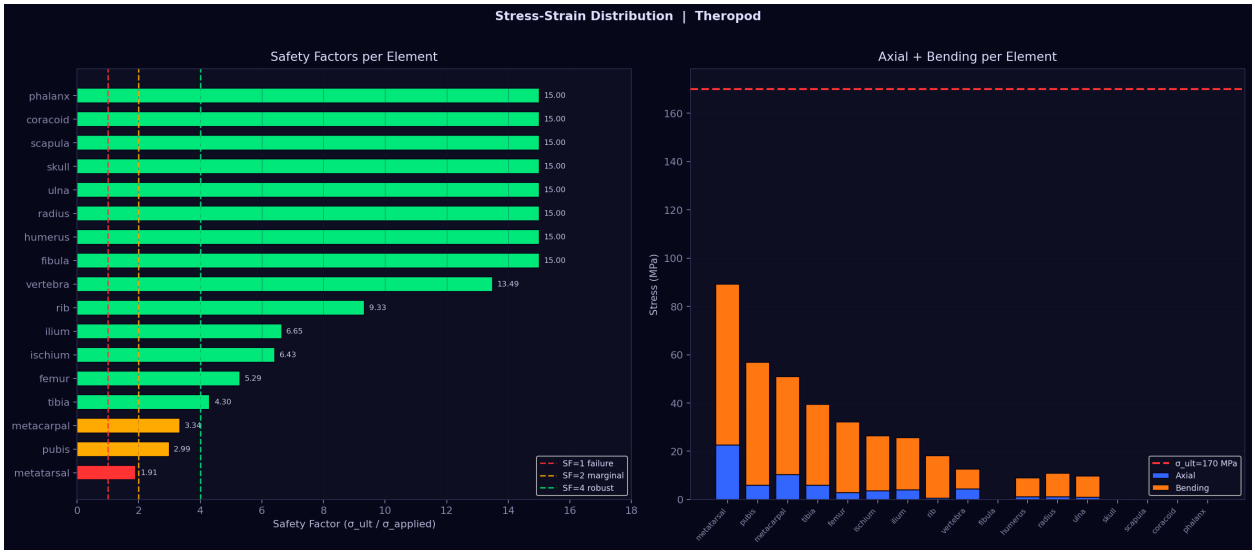
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

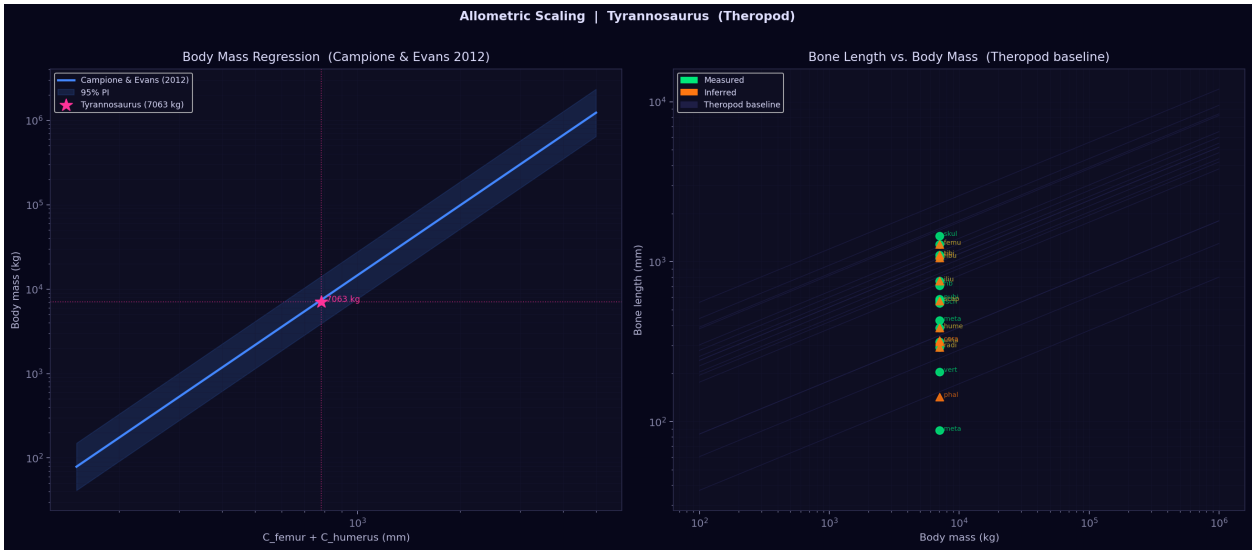


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.039	0.066	0.105	15.000	OK
femur	2.924	29.205	32.129	5.291	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	1.167	7.833	9.000	15.000	OK
ilium	3.949	21.631	25.580	6.646	OK
ischium	3.577	22.877	26.454	6.426	OK
metacarpal	10.401	40.455	50.856	3.343	WARNING
metatarsal	22.501	66.719	89.220	1.905	CRITICAL
phalanx	0.000	0.000	0.000	15.000	OK
pubis	5.840	50.950	56.790	2.993	WARNING
radius	1.145	9.752	10.897	15.000	OK
rib	0.556	17.668	18.224	9.328	OK
scapula	0.052	0.186	0.238	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	5.811	33.697	39.508	4.303	OK
ulna	1.009	8.704	9.713	15.000	OK
vertebra	4.290	8.312	12.602	13.490	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	2.806 m	Femur + tibia + metatarsal
Estimated Speed	20.0 km/h (5.55 m/s)	Gait: Trot | Fr=1.12
Stride Length	6.658 m	Alexander (1976) / Thulborn (1990)
Bite Force	46073.7 N (4696.6 kgf)	Calibrated theropod regression (Bates & Falkingham 2012)
Skull Length	1450.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.4222	Circumference / length (dimensionless)
Limb Strength Index	5.4e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	10.72 m	Allometric from femur length
Est. Neck Length	1.5 m	Allometric from femur length
Est. Tail Length	4.07 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	321.4	250.7	8091.74	10100.9	Constraint	0.40	15.00
femur	1280.0	172.0	15167.95	18934.2	Measured	0.88	5.29
femur_right	1280.0	172.0	15167.95	18934.2	Symmetry	0.84	5.29
fibula	1060.0	46.0	634.18	665.1	Measured	0.88	15.00
fibula_right	1060.0	46.0	634.18	665.1	Symmetry	0.84	15.00
humerus	385.0	77.0	914.33	1141.4	Measured	0.88	15.00
humerus_right	385.0	77.0	914.33	1141.4	Symmetry	0.84	15.00
ilium	755.0	148.0	6624.15	8268.9	Measured	0.88	6.65
ilium_right	755.0	148.0	6624.15	8268.9	Symmetry	0.84	6.65
ischium	548.0	92.0	1857.87	2319.2	Measured	0.88	6.43
metacarpal	88.0	25.0	8.21	6.8	Measured	0.88	3.34
metatarsal	428.0	62.0	659.00	822.6	Measured	0.88	1.91
phalanx	142.9	52.2	47.01	36.4	Constraint	0.40	15.00
pubis	585.0	72.0	1214.73	1516.4	Measured	0.88	2.99
radius	292.0	46.0	247.49	308.9	Measured	0.88	15.00
radius_right	292.0	46.0	247.49	308.9	Symmetry	0.84	15.00
rib	710.0	36.0	368.57	460.1	Measured	0.88	9.33
scapula	568.0	215.0	10516.84	13128.2	Measured	0.88	15.00
scapula_right	568.0	215.0	10516.84	13128.2	Symmetry	0.84	15.00
skull	1450.0	590.0	202177.16	252377.7	Measured	0.88	15.00
tibia	1098.0	122.0	6546.09	8171.5	Measured	0.88	4.30
tibia_right	1098.0	122.0	6546.09	8171.5	Symmetry	0.84	4.30
ulna	315.0	49.0	302.94	378.2	Measured	0.88	15.00
ulna_right	315.0	49.0	302.94	378.2	Symmetry	0.84	15.00
vertebra	205.0	142.0	1655.73	2066.9	Measured	0.88	13.49